

Subject Description Form

Subject Code	AAE5111
Subject Title	Human Factors, Accident Prevention and Aviation Safety Risk Management
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. the essential concepts, ideas of human factors 2. accident prevention approaches in aviation 3. the research methodology in assessing human performance and errors; and 4. the approach for safety risk management
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. relate human cognitive and physical capabilities and limitations to the design of human-machine systems in aviation; b. apply sound methods to identify and analyze sources of human errors for aviation accident prevention; c. conduct hazards analysis, incident analysis and provide design recommendations for aviation safety; and d. design human factor experiments and conduct safety evaluation via research methodology.
Subject Synopsis/ Indicative Syllabus	Human factors basics: Physical ergonomics; Cognitive ergonomics; Organizational ergonomics. Research methods: Statistical analysis, posture analysis, workload assessment technique, questionnaire/survey design, user experiment design. Accident analysis and prevention: Accident prevention management; Safety assessment, hazard identification and resolution; Swiss Cheese Model; Human Factors Analysis Classification System. Safety Risk Management: Human errors and violations; Human Reliability analysis; Safety hazards identification; Risk assessment.

Teaching/Learning Methodology	Teaching is conducted through class lectures and case study. The basic knowledge, research methodology and theoretical models will be introduced. The understanding of how to address and identify the human factors problem and formulate the resolution will be emphasized. Research methodology, case study and analytics skills are taught in class as well as the related real-life scenarios to enhance the teaching and learning abilities. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Case Study</td><td></td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Case Study		√	√	√																					
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignment</td><td>20%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Mini Project</td><td>30%</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>3. In-class Test</td><td>10%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Final examination</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.6 × Continuous Assessment + 0.4 × Final Examination The continuous assessment (60%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via individual assignment, in-class test, and mini project. In particular, an individual assignment is used to assess the students understanding of human-centered design and physical ergonomics. In-class test is used to enhance students’ engagement and discussion. Mini projects are used to assess students’ capacities of self-study, problem-solving, and effective communication skills. In addition, mini projects train students’ systematic thinking and logical thinking in addressing human factors issues in aviation. The final examination (40%) will also be considered to assess the students’ learning outcome.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Assignment	20%	√	√			2. Mini Project	30%			√	√	3. In-class Test	10%	√	√	√	√	4. Final examination	40%	√	√	√	√	Total	100%				
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Student Study Effort Expected	Class contact:	
	▪ Lecture/Case Study	39 Hrs.
	Other student study effort:	
	▪ Self-learning/preparation	36 Hrs.
	▪ Literature study/case study/reading	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. Campbell, R. D., & Bagshaw, M. (2008). Human performance and limitations in aviation. John Wiley & Sons. 2. De Florio, F. (2016). Airworthiness: An introduction to aircraft certification and operations. Butterworth-Heinemann. 3. Dekker, S. (2004). Ten questions about human error: A new view of human factors and system safety. CRC Press. 4. Kirwan, B. (2017). A guide to practical human reliability assessment. CRC press. 5. Rodrigues, C. C., & Cusick, S. K. (2012). Commercial aviation safety. McGraw-Hill Education. 6. Stolzer, A. J., Sumwalt, R. L., & Goglia, J. J. (2023). Safety management systems in aviation. CRC Press. 7. Tsang, P. S., & Vidulich, M. A. (Eds.). (2002). Principles and practice of aviation psychology. CRC Press. 8. Wiegmann, D. A., & Shappell, S. A. (2017). A human error approach to aviation accident analysis: The human factors analysis and classification system. Routledge. 9. Wise, J. A., Hopkin, V. D., & Garland, D. J. (Eds.). (2016). Handbook of aviation human factors. CRC Press.	

Nov 2023